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M3ALEM+: A smart geolocated service marketplace with real-time matching, secure payment, and AI-based fraud detection

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A B S T R A C T

The informal economy for skilled artisans remains fragmented, making it difficult for clients to locate qualified providers. This paper presents M3ALEM+, a full-stack mobile platform for geolocation-based matching between service seekers and local artisans. The platform integrates real-time request dispatching with quality-driven ranking, bidirectional WebSocket chat, AI-assisted recommendation and fraud detection, and secure escrow payment. The backend uses Django REST Framework with Django Channels, while the mobile frontend uses React Native (Expo). An AI microservice handles recommendation and anomaly detection for fraudulent activities. The system is containerised with Docker and deployable as a production-grade application.

Code metadata

Current code version	V1.0.0
Permanent link to code/repository	https://github.com/achraf-bouchra/m3alem-plus
Permanent link to reproducible capsule	N/A
Legal Code License	MIT
Code versioning system used	git
Software code languages, tools, and services used	Python 3.11, JavaScript (React Native), Django 4.2, Django REST Framework, Django Channels 4.0, MySQL 8.0, Redis 7, scikit-learn 1.3, Docker 24
Compilation requirements, operating environments and dependencies	Docker Compose 2.20+; Python >= 3.10; Node.js >= 18
If available, link to developer documentation/manual	https://github.com/achraf-bouchra/m3alem-plus/wiki
Support email for questions	achraf.bouchra@edu.uca.ma

Software metadata

Current software version	V1.0.0
Permanent link to executables of this version	N/A (source code + Docker deployment)
Legal Software License	MIT
Computing platforms / Operating Systems	Android 10+, iOS 14+ (React Native); Linux Ubuntu 22.04 LTS (backend)
Installation requirements & dependencies	Docker >= 24.0; Docker Compose >= 2.20; Android Studio 2023+ or Xcode 15+
If available, link to user manual	https://github.com/achraf-bouchra/m3alem-plus/blob/main/README.md
Support email for questions	achraf.bouchra@edu.uca.ma

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1. Motivation and significance

The on-demand local services market encompassing plumbers, electricians, carpenters, painters, and similar skilled tradespeople suffers from persistent informational asymmetries that reduce economic efficiency and trust. Clients typically rely on word-of-mouth referrals, unstructured social media queries, or fragmented web directories to locate available artisans, while service providers lack efficient digital channels to reach nearby customers in real time [1,2]. These frictions increase response latency, reduce market efficiency, and expose both parties to trust and payment risks. In developing and emerging-market contexts, particularly across the Middle East and North Africa (MENA) region, these challenges are compounded by limited penetration of formal service platforms and a predominantly cash-based informal economy [3].

Digital platform solutions for skilled trades have emerged in various forms, ranging from static web directories to appointment-booking applications. However, the majority of existing solutions lack three capabilities that are critical for a seamless user experience in time-sensitive service contexts: (i) real-time proximity-based artisan discovery and instant dispatch mechanisms comparable to those employed in ride-hailing platforms [4]; (ii) bidirectional, low-latency in-app communication that eliminates the need for external messaging applications; and (iii) automated trust mechanisms that can flag fraudulent ratings, fake reviews, or anomalous platform behaviour before they degrade service quality [5,6].

M3ALEM+ addresses these gaps by proposing an integrated mobile platform that combines intelligent geolocation matching, a real-time request lifecycle management system, an AI-powered recommendation and fraud detection service, and a secure end-to-end payment workflow. The name is derived from the Arabic term m3alem meaning master craftsman or skilled professional, reflecting the platform's focus on the Maghrebi and broader MENA informal labour context, while remaining architecturally generalisable to other regional settings and service categories.

From a scientific software perspective, M3ALEM+ constitutes a reusable reference implementation demonstrating the integration of heterogeneous modern technologies including asynchronous Django, React Native, and lightweight machine learning microservices within a coherent, deployable system. The principal contributions presented in this paper are: (1) a real-time geolocation-based service matching engine with composite quality scoring; (2) a quality-ranked artisan recommendation model incorporating collaborative filtering; (3) a behavioural anomaly and fake-review detection module based on Isolation Forest; (4) a secure in-app payment integration with escrow-style fund holding; and (5) a containerised microservice architecture ready for production deployment and academic experimentation.

2. Software description

This section presents the M3ALEM+ platform through its layered architecture, component decomposition, and the principal functionalities accessible to end users.

2.1. System architecture

M3ALEM+ is developed as a layered microservice-based platform, as illustrated in Fig. 1. The system consists of a React Native mobile application built with Expo, enabling clients and artisans to access services through a unified cross-platform interface. The frontend communicates with a Django REST Framework backend via REST APIs for business operations and WebSocket connections for real-time messaging and notifications. A dedicated real-time service based on Django Channels and Redis supports instant communication and job-status updates. Recommendation and fraud detection functionalities are deployed as an independent AI microservice accessible through lightweight REST endpoints. Persistent data storage is provided by MySQL with spatial indexing support for efficient geolocation queries. The complete platform is containerised using Docker Compose to ensure reproducible deployment across different environments.

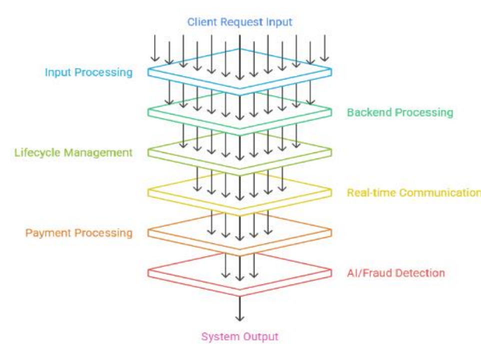

Fig. 1. System architecture of M3ALEM+

2.2. Software functionalities

Based on the proposed software architecture, the M3ALEM+ platform is designed and developed to support the complete lifecycle of home-service transactions between clients and artisans. The platform integrates geolocation-based artisan discovery, service request management, real-time communication, payment processing, and AI-assisted fraud detection functionalities. Before illustrating the operational workflow, several supporting mechanisms are implemented to facilitate platform operation, including user authentication and authorization, artisan profile verification, geospatial indexing, WebSocket-based communication, and machine-learning services for review moderation.

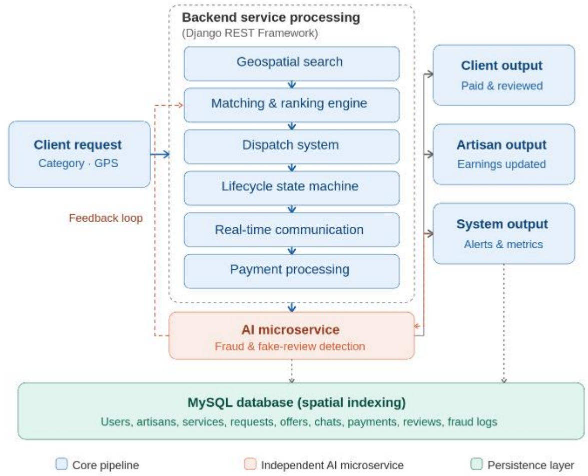

Fig. 2. M3ALEM+ platform workflow

When a client enters the platform, a service request can be created by selecting a service category, providing a textual description of the problem, and allowing the application to capture the current GPS coordinates of the device. The request information is transmitted to the backend server, where a spatial query is executed to identify artisans located within a configurable geographic radius and currently available to perform the requested service.

After retrieving the candidate artisans, the matching module applies a ranking mechanism to determine the most suitable service providers for the client request. For each candidate artisan, a matching score is calculated according to Eq. (1), where distancekm represents the geographical distance between the client and the artisan, and rating denotes the artisan's average service rating on a five-point scale. The scoring function favours artisans who are geographically closer and highly rated by previous clients. Candidate artisans are subsequently sorted in ascending order of the computed score, such that lower scores correspond to higher matching priority.

Score = distance_{km} + (5 - rating)

The resulting ranked list is then used to notify eligible artisans about the service request. Similar to on-demand service platforms, artisans may either respond by submitting a service offer or ignore the request. If no response is received within a predefined time interval, the request can be proposed to the next highest-ranked artisan until a suitable offer is obtained. This ranking mechanism enables efficient allocation of service requests while balancing geographical proximity and service quality considerations.

Upon receiving a service request, artisans may submit service offers containing the proposed service conditions and pricing information. The client is then allowed to evaluate the received offers and either accept or reject them. Once an offer is accepted, the selected artisan becomes assigned to the request, while all remaining offers associated with the same request are automatically rejected. This mechanism guarantees that only one artisan can be assigned to a service request at any given time.

Following offer acceptance, a dedicated communication channel is automatically established between the client and the artisan. The real-time communication module is implemented using Django Channels and WebSocket technology, allowing participants to exchange messages instantly throughout service execution. All messages are persisted in the database and synchronised through Redis-based channel layers. In addition to user-generated messages, the communication module distributes notifications related to service status updates and system events.

The service request itself follows a predefined lifecycle managed by the backend system. Initially, newly created requests are assigned the PENDING state. Once artisans become eligible to submit offers, the request enters the OPEN state. Following offer acceptance, the request transitions to the ASSIGNED state and remains active until service completion. After successful execution and payment confirmation, the request is marked as DONE. Additional states such as CANCELLED are supported to handle exceptional situations. All state transitions are recorded in the database to ensure traceability and support auditing procedures.

After the requested service has been completed, the payment workflow is initiated. The platform creates a payment transaction associated with the accepted offer, the client, and the assigned artisan. Transaction information includes the payment amount, selected payment method, transaction status, and fraud analysis results. Similar to the request lifecycle, payments progress through predefined states, namely PENDING, PAID, FAILED, REFUNDED, and FRAUD. Upon successful payment validation, the corresponding service request is automatically finalised and recorded as completed.

To improve platform trustworthiness, the M3ALEM+ platform incorporates an AI-assisted fraud detection module operating independently from the core application services. The fraud detection process primarily focuses on review authenticity analysis and suspicious behavioural patterns. When a client submits a rating or textual review, the content is transmitted to the AI microservice for analysis. The system computes a fraud probability score, a fake-review classification result, and a moderation status based on machine-learning predictions and rule-based validation mechanisms.

Reviews identified as potentially fraudulent are flagged for moderation and may be temporarily excluded from artisan reputation calculations until verification is completed. Fraud-related events are additionally stored as platform alerts to facilitate administrative investigations and maintain marketplace integrity. By combining automated machine-learning analysis with human moderation procedures, the platform contributes to maintaining trust, fairness, and reliability within the digital marketplace ecosystem.

3.3. Dataset description

3.3.1. Dataset for fake review detection

The fraud detection component relies on a curated dataset of 71,885 user-generated reviews collected and preprocessed in 2023. The dataset covers local services and artisan activities in French and English, with four main features: category (service type), rating (1-5), label (0 = genuine, 1 = fake/spam), and text (review content).

Property	**Value**
Number of reviews	71,885 entries
Main features	category, rating, label, text
Target class	0 (genuine), 1 (fake/spam)
Domains	Local services, artisans, small businesses
Language	French / English
Collection year	2023

Table 1. Characteristics of the Fake Review

3.3.2. Analytical objectives

The dataset supports binary classification of reviews, detection of inconsistencies between textual sentiment and numeric ratings, and identification of suspicious patterns including repetitive content, coordinated rating bursts, and abnormal user activity profiles.

3.3.3. Integration within M3ALEM+

The dataset is integrated into the AI microservice, where scikit-learn-based models are trained to detect fraudulent reviews in real time. This mechanism enhances trust and ensures the reliability of user-generated content within the platform ecosystem.

4. Illustrative examples

In this section, the operational processes of the proposed M3ALEM+ platform are illustrated through the client and artisan workflows, as shown in the corresponding figures, where only the essential user interfaces are presented.

To initiate a service request, the client opens the M3ALEM+ mobile application and selects a service category such as plumbing from the service taxonomy. The application automatically captures the device GPS coordinates and submits a service request containing the user's location, a free-text description of the problem, and an optional image attachment.

The backend system processes the request by performing a spatial query to identify available artisans within a configurable geographic radius. A composite scoring function is then applied to rank candidates based on distance, service rating, experience, and availability status. The system dispatches real-time notifications to the highest-ranked artisans.

Upon receiving a request, the selected artisan may respond by submitting a service offer that includes availability and proposed service terms. The client then evaluates the received offer and may either accept or reject it. Once an offer is accepted, both parties are notified, and a dedicated real-time communication channel is automatically established to enable in-app chat and status updates. Optionally, the client can track the artisan's live location during transit to the service location.

Once the service is completed, the system triggers the payment workflow through the integrated payment gateway. After successful payment confirmation, the request status is updated to completed, and the client is allowed to submit a rating and review for the provided service.

In parallel, artisans register on the platform by providing identity credentials, selecting one or more professional categories, defining their service area using a radius or polygon-based mapping interface, and uploading verification documents. Following successful verification through a KYC process, the artisan profile becomes active and eligible for dispatch.

When marked as available, artisans can receive incoming service requests and appear in client search results. The artisan dashboard provides performance indicators including average rating, number of completed jobs, acceptance rate, response time, and a composite ranking score, which are used by the matching algorithm to optimise dispatch decisions. Earnings are continuously tracked, and artisans can request withdrawals through supported payment methods such as bank transfer or mobile wallets.

To ensure platform trustworthiness, a fraud detection module continuously monitors user activity and review patterns. In a simulated fraud scenario, malicious users attempt to manipulate ratings by submitting multiple high-rated reviews from newly created accounts within a short time window. The system extracts behavioural and textual features such as review frequency, account age, text similarity, and absence of validated transactions. An Isolation Forest-based anomaly detection model is then applied to identify suspicious patterns. Detected fraudulent reviews are suppressed from public visibility, forwarded to a manual moderation queue, and logged for auditing purposes. The artisan's official rating remains unchanged until verification is completed, ensuring fairness in ranking and dispatch prioritisation.

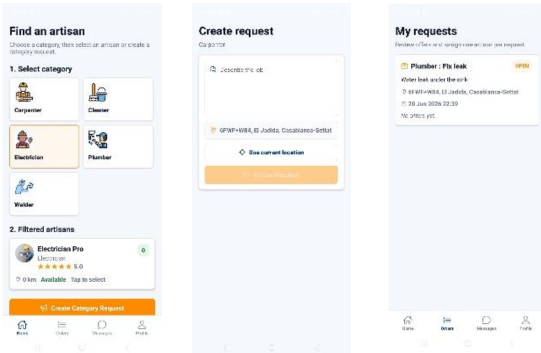

Fig. 3. Workflow of request & matching

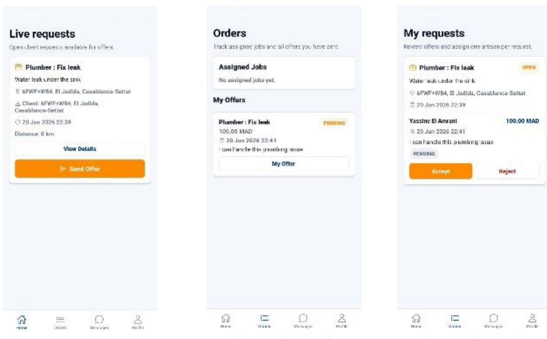

Fig. 4. Workflow of offer

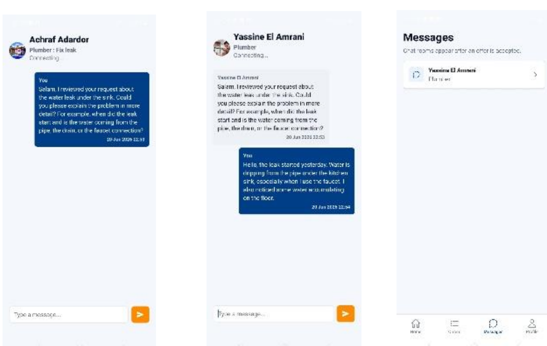

Fig. 5. Chat between client and artisan

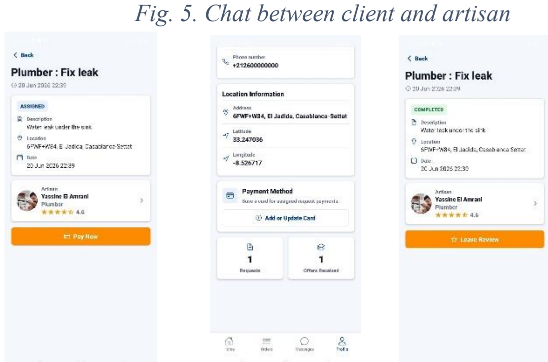

Fig. 6. Payment interface

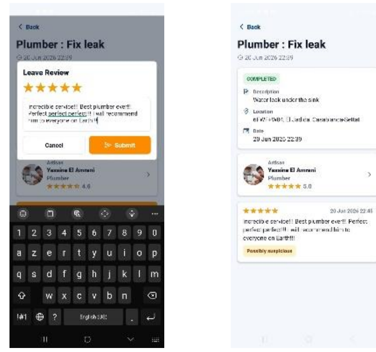

Fig. 7. Fraude detection interface

6. Comparison with existing platforms

The Moroccan digital services landscape includes several platforms connecting clients with artisans, among which Maalam.ma represents one of the most established solutions. While Maalam.ma laid the foundation for digitising client–artisan interactions in Morocco, it remains limited by a static and manually driven workflow, where service requests are processed asynchronously with response delays ranging from hours to days, communication is handled externally via phone calls, artisan listings are presented without quality-based ranking, no automated fraud detection mechanisms are implemented, and payments are conducted offline without transaction security or traceability.

In contrast, M3ALEM+ is designed to overcome these limitations by introducing a real-time, intelligent service model that enables instant dispatch of nearby artisans, integrated in-app communication, algorithmic ranking based on multiple quality indicators, AI-driven fraud detection, and secure escrow-style payment processing, providing an end-to-end digital service experience.

Feature	Maalam.ma	M3ALEM+
Real-time matching	Not available (manual quotes)	Instant dispatch with cascading offers
In-app communication	External phone/messaging required	Integrated WebSocket chat
Artisan ranking	Chronological listing	Composite quality scoring + collaborative filtering
Fraud detection	None	Isolation Forest anomaly detection
Payment processing	Cash only (external)	Escrow-style secure payment
Mobile application	Web-only	React Native (iOS + Android)
Geolocation support	City-level filtering	GPS-based proximity queries
AI recommendation	None	Hybrid collaborative + content-based filtering

Table 2. Comparison between Maalam.ma an M3ALEM+

7. Reusability and impact

Overall, M3ALEM+ is designed with modularity and extensibility as core architectural principles. The AI microservice is fully decoupled from the main backend and exposed through a versioned REST interface, allowing independent deployment, scaling, and replacement of recommendation and fraud detection models without affecting the core system. Model retraining can be executed either on a scheduled or on-demand basis, ensuring continuous improvement of AI components without service disruption.

The platform further adopts a dynamic, database-driven service taxonomy, enabling the extension of artisan categories to new domains such as healthcare support, private tutoring, or logistics services without requiring code-level modifications. This design enhances system adaptability and supports long-term evolution across diverse service ecosystems.

From a research and development perspective, M3ALEM+ is structured as a reference implementation for location-based service platforms and gig-economy systems. The containerised deployment using Docker Compose ensures reproducibility across academic, development, and production environments, facilitating experimentation, evaluation, and system replication. In addition, environment-based configuration management improves deployment security and portability by avoiding hard-coded credentials or system-specific parameters.

From a socio-economic perspective, the platform contributes to the digital transformation and gradual formalisation of informal skilled labour markets, particularly in developing regions. By providing artisans with structured demand, reputation tracking, and secure payment mechanisms, M3ALEM+ improves income transparency, strengthens trust between users, and enhances accountability in service delivery. The integrated fraud detection module further reinforces platform reliability by mitigating reputation manipulation and ensuring fair ranking of service providers.

Future enhancements include multilingual support for Arabic, French, and Tamazight to improve regional accessibility, integration of real-time artisan tracking during service execution, and expansion of the AI module with natural language processing for sentiment analysis and review authenticity verification. Additional directions also include subscription-based premium visibility features for artisans and integration with external mobility services to improve arrival time estimation in urban contexts.

8. Conclusion

In summary, M3ALEM+ is designed and developed as a smart geolocated service marketplace to facilitate on-demand interactions between clients and artisans. The platform integrates real-time artisan discovery, service request management, instant communication, secure payment processing, and AI-assisted fraud detection within a unified mobile environment. To improve the efficiency and trustworthiness of local service transactions, a ranking mechanism based on geographical proximity and service ratings is employed to identify suitable artisans, while WebSocket technology enables real-time communication throughout service execution. In addition, machine-learning techniques are incorporated to detect suspicious reviews and behavioural anomalies, contributing to platform reliability and user confidence. By combining location-based matching, intelligent service recommendation, fraud prevention, and secure transaction management, M3ALEM+ provides a practical solution for digitalising local service ecosystems and can serve as a reference platform for future research and development in mobile marketplaces, location-based services, and AI-enabled service platforms.

CRediT authorship contribution statement

Achraf: Conceptualization, Methodology, Software, Writing – original draft, Visualization.

Bouchra: Software, Validation, Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have influenced the work reported in this paper.

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Data availability

Data and source code will be made available upon reasonable request to the authors.

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